Analysis of machine learning models for glass Identification

Introduction

The aim of this project is to identify glass types based on their chemical material using K-nearest Neighbour (KNN), Decision Trees (DT) and Naïve Bayes (NB) machine learning models to determine the best classification performance for real world investigations and legalities. Accurate identification is important because different types of glass are designed for specific purpose. For example, in a car accident identifying whether the correct type of glass was used could determine legal responsibilities, while in lamps, incorrect glass identification could lead to breakage of the lamp, due to heat.

Data and Preparation

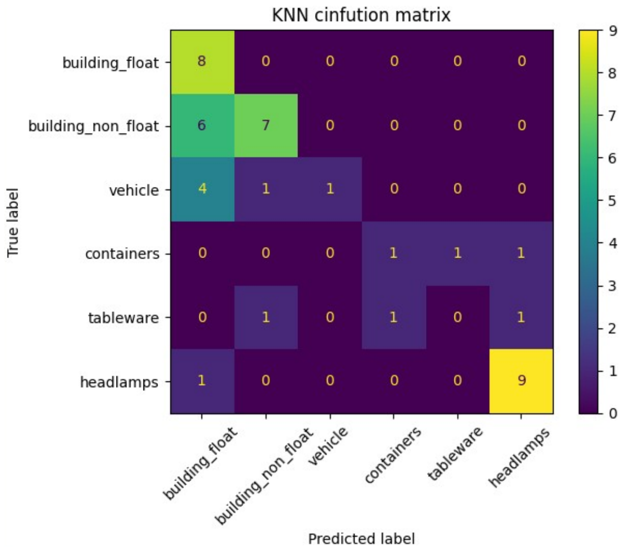
The class identification dataset contains 214 samples with 9 features containing chemical properties such as reflective index and chemical content. Each sample is categorised into one of several glass types. Initially before processing the dataset the ID attribute was removed and feature values were normalised using Min-Max scaling to make sure all features have equal contribution to the distance based calculation of KNN. The data was then split with the ratio of 8:2 for training and testing purposes respectively, using a random seed of student ID to ensure reproducibility.

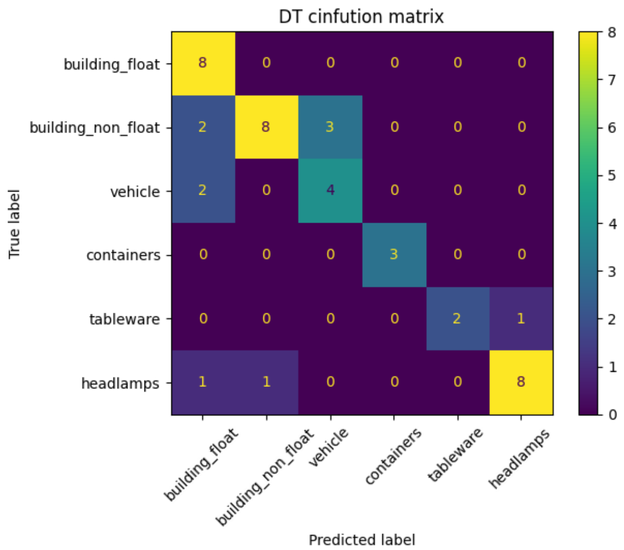
Methodology

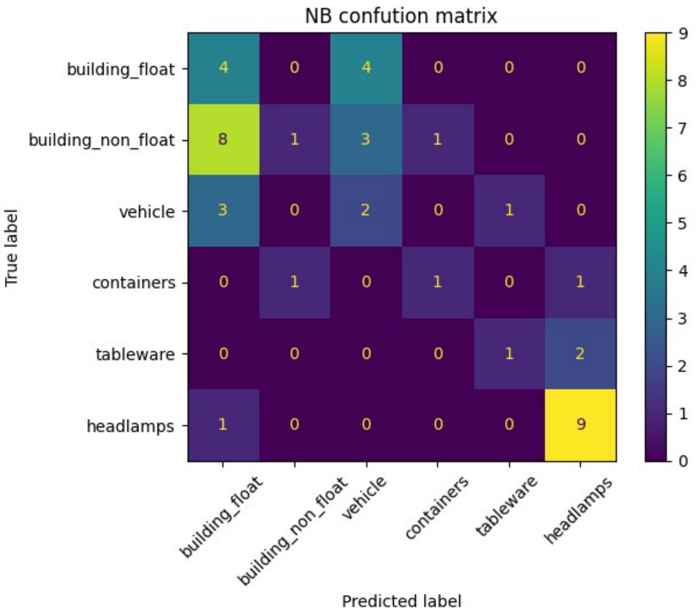
Three machine learning models were implemented from scratch: K-nearest Neighbour (KNN), Decision Trees (DT) and Naïve Bayes (NB). KNN identifies samples based on the majority classes of its nearest neighbours using Euclidean distance. The value of K was decided using trial and error. The DT recursively split the data using entropy and information gain to find the best feature splits, creating a tree structure for classification. The NB algorithm applied Bayes theorem with Gaussian distribution to calculate class probabilities based on the distribution of the features, while assuming feature independency. Performance was measured using accuracy, training time, testing time and confusion matrix. Models were chosen because they represent distance based (KNN), tree based (DT) and probability based (NB) algorithms allowing a direct comparison of strength.

Results

Model	Accuracy	Training time/s	prediction time/s
KNN(k=6)	60.4%	~0	0.000517
DT	76.7%	0.2009	0.000065
NB	41.8%	0.0007	0.001746







The DT achieved the highest accuracy of 76.7% and the fastest prediction time, correctly identifying all samples for class 1 and performing well on class 2 and 7. KNN delivered a average accuracy of 60.4% with significant confusion in class 2, KNN also required more computational power during testing. NB performed worst, with accuracy of 41.8% misidentifying classes 1-3 due to its strong feature independence while being the fastest to train.

Conclusion

The DT model is recommended for glass identification. It delivered the highest accuracy with significant prediction speed. Although it has a high training cost, it is acceptable given its performance, making it an excellent choice for investigation purposes where both accuracy and speed matters. Benefits include good management of non linear relationships and class variation without requiring large processing power. Limitations include longer training time and potential vulnerability to over fitting on small databases. Further work could explore enhancing techniques like pruning for DT and optimal classification trees to improve performance.